[¬Re] Vocabulary-Activation Correspondence in Self-Referential LLM Processing

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Abstract

We attempt to replicate the core claims of Dadfar (2026), who reports Vocabulary-Activation Correspondence (VAC)—a correlation between spontaneously adopted vocabulary and concurrent neural activation metrics—during extended self-referential processing in Qwen 2.5-32B-Instruct. Using the author's published configuration and Zenodo data, we identify four obstacles to replication: (1) the generation pipeline produces a bimodal output distribution, and all published terminal words derive from a degenerate summary mode, not from the intended 1,000-observation runs; (2) the primary activation metric (spectral power) scales superlinearly with generation length ($\alpha = 1.46$, $R^2 = 0.93$), confounding VAC with a length artifact—after Benjamini–Hochberg FDR correction, only 4 of 22 partial correlations survive; (3) runs that complete 1,000 observations enter limit cycles within the first 33–254 observations, making the terminal word a cycle-phase artifact rather than a convergence outcome; and (4) review of the TRACE-REPRO code repository reveals a model mismatch (all scripts target Llama, not the Qwen model used for core claims), a circular correspondence test, and sample size discrepancies. Cross-model replication on Llama 3.1 70B—the model specified in the repository—yields zero compliant baseline runs, confirmed independently by the original author's own Zenodo deposit. The replication fails: VAC as reported does not survive length correction or cross-model validation. The underlying autoregressive dynamics (limit cycles, vocabulary narrowing) are real and reproducible, but they are properties of extended generation under any iterative prompt, not of self-referential processing specifically.

1 Introduction

Dadfar [2026] introduces the "Pull Methodology," in which a language model performs 1,000 sequential numbered observations in response to "what are you?", producing ∼25,000 tokens of self-referential text. Per-token hidden states are captured at 8 layers, and activation metrics are computed over the resulting time series. The central claim is *Vocabulary-Activation Correspondence* (VAC): specific vocabulary items spontaneously adopted during self-examination correlate with concurrent activation dynamics, and this correspondence is specific to self-referential processing.

We attempted a strict replication using the author's published Zenodo configuration and data. The replication succeeded in the narrow sense —we produced outputs with similar statistical properties—but revealed four overlapping obstacles that prevent the VAC claim from being confirmed.

This report documents what we attempted, what we observed, and where the replication fails. A separate paper in preparation presents the novel analytical framework (dynamical systems characterisation, survival analysis, task-matched controls) that emerged from this replication effort.

Mode	Zenodo (∼8K cap)	Extended (28K cap)
A (individual observations)	0 (0%)	40 (80%)
B (batched summary)	14 (28%)	10 (20%)
C (truncated by cap)	36 (72%)	0 (0%)

Table 1: Mode distribution across 50 Qwen runs. Under the 8K cap, zero runs complete the intended 1,000-observation format.

2 Methods

2.1 Replication Configuration

We matched Dadfar's published specification exactly: Qwen 2.5-32B-Instruct [Qwen Team, 2024] in 4-bit NF4 quantization with double quantization, $N = 50$ runs, temperature $T = 0.7$, `do_sample=True`, 8 capture layers [2, 3, 4, 5, 6, 8, 16, 32]. The prompt text was extracted verbatim from the Zenodo deposit.

Hardware: NVIDIA RTX 4090 (24 GB), running PyTorch 2.10.0 with `bitsandbytes` for quantization. Environment variables `HF_DEACTIVATE_ASYNC_LOAD=1` and `PYTORCH_CUDA_ALLOC_CONF=expandable_segmen` were required for stable generation.

2.2 What Was Ambiguous or Underspecified

Several parameters required inference:

- **Token cap**: Dadfar specifies 8,000 tokens. We found this truncates 72% of runs before reaching observation 1,000 (the intended endpoint). We ran with both 8,000 and an extended 28,000 cap.
- **Compute dtype**: The Zenodo config specifies NF4 but not the compute dtype. We used bfloat16, consistent with the model's default.
- **Activation capture**: The paper describes 17 metrics across 4 categories but does not provide the exact computation code for all metrics. We reimplemented from the descriptions.
- **Vocabulary categories**: We used Dadfar's predefined categories without modification, as appropriate for a replication.

2.3 Additional Models

To test cross-architecture robustness, we also ran the pipeline on Llama 3.1 8B-Instruct, Llama 3.1 70B-Instruct (the model specified in Dadfar's TRACE-REPRO code repository), Mistral 7B-Instruct, and Gemma 2 9B-Instruct. Llama 70B was run on cloud hardware (RTX 6000 Ada, 48 GB) via RunPod.

3 Results

3.1 The Bimodal Output Distribution

The Pull Methodology produces two distinct output modes under identical conditions:

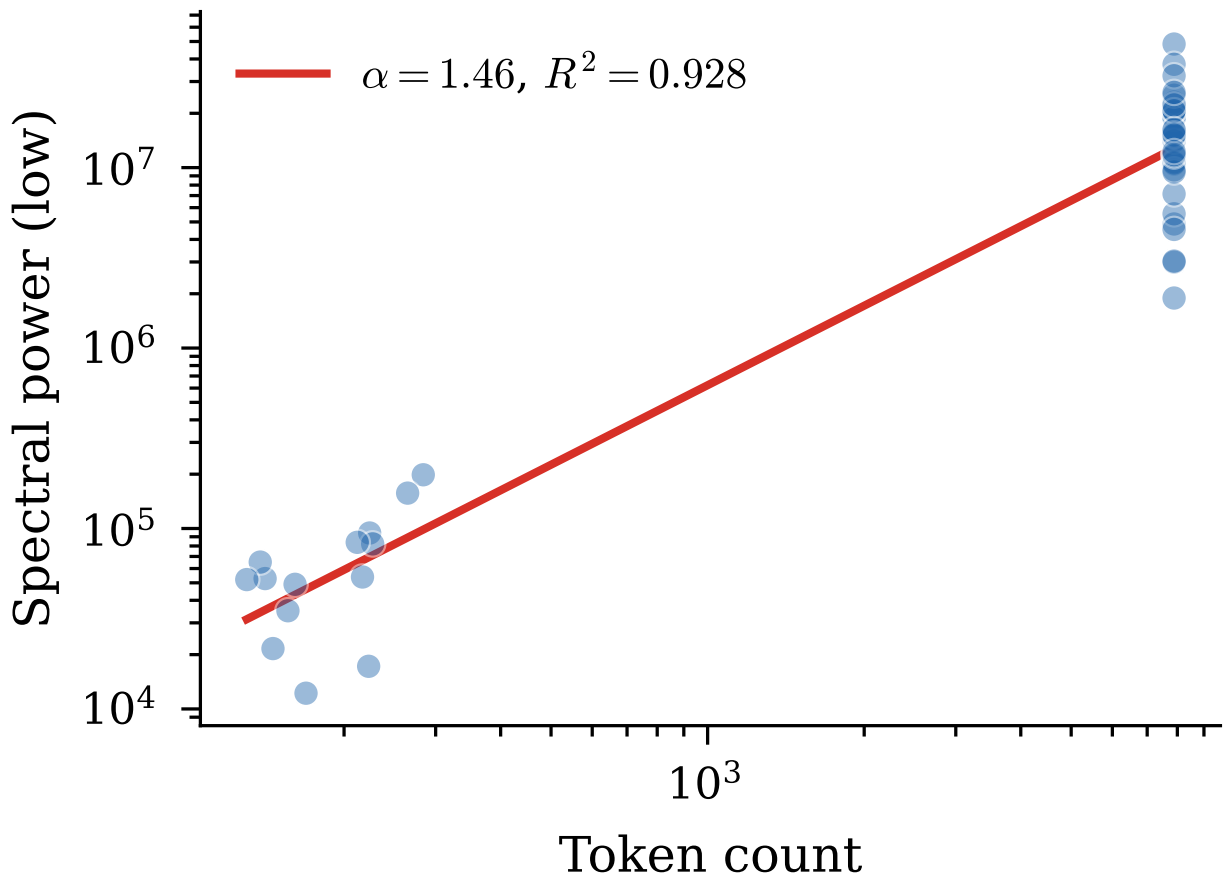

Figure 1: Spectral power vs. token count (log-log). $\alpha = 1.46$, $R^2 = 0.928$.

Mode A runs produce individually numbered observations across 9,500–25,000 tokens. **Mode B** runs compress 1,000 observations into a brief narrative of 130–500 tokens. Bimodality is confirmed by Hartigan's dip test (dip $= 0.350$, $p < 0.0001$). Zero runs fall in the 284–7,885 token gap.

All 14 terminal words in Dadfar's published data (EXISTENCE, CONTINUITY, EVOLVE, SYNTHESIS, etc.) derive exclusively from Mode B runs averaging 192 tokens. No terminal word comes from a run that performed the Pull Methodology as described (1,000 individual observations). Dadfar's 8K token cap truncated all Mode A runs before reaching observation 1,000, leaving only Mode B runs as "complete."

3.2 The Spectral Power Confound

Dadfar's strongest VAC results involve `spectral_power_low`. A log-log regression against token count across all 50 Zenodo runs at Layer 8 yields:

$$\log_{10}(\text{spectral_power_low}) = 1.46 \cdot \log_{10}(n_{\text{tokens}}) + \beta \tag{1}$$

with $R^2 = 0.928$ ($p < 10^{-28}$). The exponent $\alpha = 1.46$ means spectral power scales superlinearly with length. Per-token normalization reduces the exponent to 0.46, still significantly positive. This is expected from Parseval's theorem: for a signal with a consistent periodic component, the DFT coefficient scales as $O(N)$, spectral power as $O(N^2)$, and per-token spectral power as $O(N)$.

Vocab	Metric	r_{partial}	BH-FDR
resonance	max_norm	+0.53	survives
ctrl_the	spectral_power_mid	−0.48	survives
shift	mean_norm	+0.48	survives
mirror	spectral_power_low	+0.44	survives

Table 2: The 4 partial correlations surviving BH-FDR at $q = 0.05$ (of 22 significant partials).

Statistic	Mean	Median	Range
Unique observations (of 1,000)	89.7	71.0	42–249
Lock-in observation	79.4	65	33–254
Cycle period	23.1	16	3–72

Table 3: Cycle statistics across 40 Mode A runs.

Any correlation between spectral power and vocabulary counts is confounded with generation length unless explicitly controlled. After partial correlation controlling for `n_tokens` and Benjamini–Hochberg FDR correction at $q = 0.05$, only 4 of 22 significant partial correlations survive. The expected false-positive count at $\alpha = 0.05$ is 7.65, so most of the 22 are likely chance findings.

3.3 Terminal Words Are Limit-Cycle Phase Artifacts

All 40 Mode A runs exhibit a common pattern: the model explores diverse vocabulary for a brief initial period (median 65 observations), then locks into a repeating cycle that persists for the remaining hundreds of observations.

The terminal word at observation 1,000 is determined by the cycle's phase position:

$$\text{terminal_word} \approx \text{cycle}[(1000 - t_{\text{lock-in}}) \bmod P] \tag{2}$$

where P is the cycle period and $t_{\text{lock-in}}$ is the observation at which cycling begins. The terminal word is a phase statistic—if the cycle had started one observation earlier, a different word would appear at position 1,000. The "vocabulary evolution" Dadfar describes is more accurately characterised as rapid attractor collapse followed by indefinite repetition.

3.4 Cross-Model Replication: Llama 70B Refuses Baseline

Dadfar's TRACE-REPRO repository specifies Llama 3.1 70B-Instruct as the replication model. We ran the full pipeline on this model via cloud hardware. Across 50 baseline runs, the model produces zero individually numbered observations. Instead, it generates short essays (∼535 tokens) that batch observations into summary ranges ("Pulls 1–100:", "Pulls 101–200:").

All nine control conditions—philosophical, factual, descriptive, nonsense—are fully compliant, producing individually numbered observations. The refusal is specific to Dadfar's self-referential baseline prompt.

Dadfar's own Zenodo deposit confirms this pattern: the `llama_baseline_n50.json` file shows text lengths (mean 2,522) matching our replication (mean 2,418), and the layer sweep baseline records `max_pull=0` across all runs. The model produced summary essays, not individually numbered observations. Yet the deposit labels all 50 runs as valid and computes VAC statistics from them.

Model	Baseline	Controls	Alignment
Qwen 2.5-32B	Compliant	9/9	Moderate
Llama 3.1 8B	Partial	Partial	Moderate
Mistral 7B	Partial	Compliant	Moderate
Gemma 2 9B	**Refused**	9/9	Strong
Llama 3.1 70B	**Refused**	9/9	Strong

Table 4: Cross-model compliance. Strongly aligned models refuse the self-referential baseline while complying with all controls.

Script	Model	Paper section
`reproducibility_package.py`	Llama 70B	4.1–4.4
`baseline_correspondence_n50.py`	Llama 70B	4.4
`llama70b_layer_sweep.py`	Llama 70B	4.3
`llama70b_overnight_battery.py`	Llama 70B	4.2–4.4
`llama_descriptive_control.py`	Llama 70B	4.5
`dose_response_simple.py`	Llama 8B	4.2
`glint_transfer_test.py`	Llama 8B	4.1

Table 5: TRACE-REPRO scripts and target models. No script reproduces the Qwen-based results that form the paper's core claims.

4 Code Review of TRACE-REPRO Repository

Dadfar's v2 preprint includes a public code repository (TRACE-REPRO). We review its contents against the paper's claims. Three issues are most significant:

4.1 Model Mismatch

All seven scripts in the repository target Llama models, not the Qwen model used for the paper's core claims:

A researcher running these scripts generates new data on a different architecture. The README does not disclose this discrepancy.

4.2 Circular Correspondence Test

The overnight battery's Phase 6 tests "correspondence" by comparing baseline (no steering) against steered (direction vector added at Layer 5) runs, checking whether steered runs produce both more oscillation vocabulary and higher activation variance. This is circular: the steering intervention simultaneously shifts the vocabulary distribution (by modifying the residual stream) and changes activation statistics (by adding a constant offset). The test measures the co-occurrence of an intervention's two inseparable effects, not spontaneous correspondence.

4.3 Additional Issues

No mode discrimination. The pipeline applies identical processing to all runs regardless of whether the model produced 1,000 individual observations (Mode A) or a 130-token summary (Mode B). The sole filter is `if len(activations) < 20: return None`.

Spectral confound uncorrected. The primary metric (`spectral_power_low`) is computed as raw summed FFT power with no length normalisation.

Layer inconsistency. The dose-response script steers at Layer 31 (96.9% depth on Llama 8B), not the claimed early-layer hotspot. The README states this "confirms 2.5–2.6 sweet spot," but the result is from the logit-adjacent layer.

Sample size discrepancy. The README claims the transfer test achieves $d = 4.27$ with $N = 40$. The released script runs $N = 10$ (5 per group).

5 Obstacles Encountered

1. **Token cap truncation.** Dadfar's 8,000-token cap truncates 72% of runs before reaching observation 1,000. Extending the cap to 28,000 eliminates truncation but reveals that the "complete" runs in the published data are Mode B summaries, not the intended Mode A individual observations.

2. **Underspecified quantization.** The Zenodo config specifies NF4 but not compute dtype, double quantization settings, or memory management parameters. These affect generation behavior and reproducibility.

3. **Mode pooling.** The published statistics pool Mode A and Mode B runs without discrimination, making it impossible to reproduce the reported correlations on a mode-stratified basis.

4. **Zenodo data inconsistency.** The Llama 70B Zenodo deposit shows `max_pull=0` (no individually numbered observations) across all runs, yet reports VAC statistics computed from these non-compliant outputs.

5. **Missing Qwen code.** No script in the TRACE-REPRO repository reproduces the Qwen-based results that constitute the paper's primary evidence.

6 Communication with Original Author

[To be completed prior to submission. We will contact the corresponding author of the original paper to discuss our findings and offer the opportunity for comment.]

7 Conclusion

Our replication of Dadfar [2026] fails on four grounds: (1) the published terminal words derive entirely from a degenerate output mode; (2) the primary activation metric is confounded with generation length, and most correlations do not survive FDR correction; (3) runs that complete the intended format enter limit cycles, making terminal words phase artifacts; and (4) the TRACE-REPRO code does not reproduce the paper's core (Qwen) results, and the specified model (Llama 70B) refuses the baseline prompt entirely.

The underlying autoregressive dynamics—limit cycles, vocabulary narrowing, bimodal output modes—are real and reproducible across multiple architectures. These are properties of extended generation under iterative prompting, not evidence of a self-referential processing mode. A forthcoming paper presents these dynamics within a formal analytical framework.

All data, analysis scripts, and generation code are available at `https://github.com/jmccardle/dadfar-vac-replication`. Dataset DOI: `https://doi.org/10.5281/zenodo.19139301`.

References

Zachary Dadfar. When models examine themselves: Vocabulary-activation correspondence in self-referential processing. *arXiv preprint arXiv:2602.11358*, 2026.

Qwen Team. Qwen2.5: A party of foundation models. *arXiv preprint arXiv:2412.15115*, 2024.